FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX | Paper: SSC-I

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

Total Marks: 65 **Time Allowed: 2 Hours 30 Minutes** **Version: 4**

Atomic Masses (amu): H=1 · C=12 · N=14 · O=16 · Na=23 · Mg=24 · Al=27 · S=32 · Cl=35.5 · Ca=40 · Cr=52 · Mn=55 · Fe=56 | $N_A = 6.022 \times 10^{23}$

General Instructions:

1. The paper has three sections: **A** (Multiple Choice), **B** (Short Questions) and **C** (Long Questions). Section A is compulsory.
2. In Section A, encircle or fill the bubble of the correct option on the question paper itself. All 12 questions are compulsory.
3. In Sections B and C, for each question attempt **either the main question OR its alternative**, not both.
4. Show **all working** for every numerical in Sections B and C. Marks are awarded for the method, not only the final answer. Use black or blue ink; pencil for diagrams only.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Which branch of chemistry deals with the manufacture of chemicals and materials on a large, industrial scale?	Analytical chemistry	Industrial chemistry	Organic chemistry	Physical chemistry	Ⓐ Ⓑ Ⓒ Ⓓ
2	Which of the following is a **homogeneous** mixture?	Muddy water	Sand in water	Salt dissolved in water	Oil and water	Ⓐ Ⓑ Ⓒ Ⓓ
3	An element has two isotopes of masses 20 amu (abundance 90%) and 22 amu (abundance 10%). Its relative atomic mass is:	20.2 amu	21.0 amu	20.0 amu	21.8 amu	Ⓐ Ⓑ Ⓒ Ⓓ
4	According to the Heisenberg uncertainty principle, it is impossible to determine, at the same time, an electron's exact:	mass and charge	position and future path	number of protons and neutrons	shell and sub-shell	Ⓐ Ⓑ Ⓒ Ⓓ
5	In the periodic table, the lanthanides and actinides belong to the:	s-block	p-block	d-block	f-block	Ⓐ Ⓑ Ⓒ Ⓓ
6	Dipole-dipole forces of attraction act between molecules that are:	non-polar	polar	monatomic	held by ionic bonds	Ⓐ Ⓑ Ⓒ Ⓓ
7	The number of moles present in 9.2 g of sodium is: (Na = 23)	0.2 mol	0.4 mol	0.5 mol	2.0 mol	Ⓐ Ⓑ Ⓒ Ⓓ
8	The number of molecules present in 0.25 mol of water (H_2O) is:	1.51×10^{23}	6.022×10^{23}	3.01×10^{23}	2.41×10^{24}	Ⓐ Ⓑ Ⓒ Ⓓ
9	The oxidation number of manganese in manganese sulphate ($MnSO_4$) is:	+2	+4	+6	+7	Ⓐ Ⓑ Ⓒ Ⓓ
10	In a reaction the sum of the bond energies broken is 690 kJ and the sum of the bond energies formed is 810 kJ. The enthalpy change (ΔH) is:	+120 kJ	−120 kJ	+1500 kJ	−1500 kJ	Ⓐ Ⓑ Ⓒ Ⓓ
11	When a reversible reaction reaches equilibrium in a closed container, the concentrations of the reactants and products:	become exactly equal	keep changing rapidly	remain constant	fall to zero	Ⓐ Ⓑ Ⓒ Ⓓ
12	As we move **down** Group 1 (the alkali metals), the reactivity of the elements:	decreases	increases	remains constant	first increases then decreases	Ⓐ Ⓑ Ⓒ Ⓓ

SECTION B — Short Questions (11 × 3 = 33 Marks)

*Attempt all 11 questions. For each, choose either the main question **OR** its alternative.*

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Calculate the number of moles present in 22.2 g of calcium chloride ($CaCl_2$). (Ca = 40, Cl = 35.5)	**3**	**OR**	Calculate the number of moles present in 5.6 g of nitrogen gas (N_2). (N = 14)	**3**
2(ii)	Calculate the number of molecules present in 0.4 mol of carbon dioxide (CO_2). ($N_A = 6.022\times10^{23}$)	**3**	**OR**	Calculate the number of atoms present in 0.3 mol of iron (Fe). ($N_A = 6.022\times10^{23}$)	**3**
2(iii)	Calculate the molar mass of calcium nitrate, $Ca(NO_3)_2$. (Ca = 40, N = 14, O = 16)	**3**	**OR**	Calculate the formula mass of ammonium sulphate, $(NH_4)_2SO_4$. (N = 14, H = 1, S = 32, O = 16)	**3**
2(iv)	Calculate the percentage by mass of nitrogen in ammonium nitrate, NH_4NO_3. (N = 14, H = 1, O = 16)	**3**	**OR**	Calculate the percentage by mass of oxygen in magnesium oxide (MgO). (Mg = 24, O = 16)	**3**
2(v)	Determine the oxidation number of the underlined atom, showing your working: (a) $\underline{S}$ in SO_3 (b) $\underline{Cr}$ in CrO_3 (c) $\underline{P}$ in P_2O_5.	**3**	**OR**	Determine the oxidation number of the underlined atom, showing your working: (a) $\underline{N}$ in N_2O_5 (b) $\underline{Cl}$ in Cl_2O (c) $\underline{I}$ in HIO_3.	**3**
2(vi)	For the reaction $2Al + 3Cl_2 \rightarrow 2AlCl_3$, calculate (a) the moles of Cl_2 needed to react completely with 0.4 mol of Al, and (b) the moles of $AlCl_3$ formed.	**3**	**OR**	In the decomposition $CaCO_3 \rightarrow CaO + CO_2$, calculate the mass of calcium oxide (CaO) produced from 20 g of calcium carbonate. (Ca = 40, C = 12, O = 16)	**3**
2(vii)	An element M has two isotopes: mass 63 amu (abundance 69%) and mass 65 amu (abundance 31%). Calculate the relative atomic mass of M.	**3**	**OR**	An element R has two isotopes: mass 10 amu (abundance 20%) and mass 11 amu (abundance 80%). Calculate the relative atomic mass of R.	**3**
2(viii)	State the **Heisenberg uncertainty principle**. What does it tell us about drawing a fixed path (orbit) for an electron around the nucleus?	**3**	**OR**	What is meant by the **quantum mechanical model** of the atom? Define an *orbital*, and state what Louis de Broglie proposed about electrons in 1924.	**3**
2(ix)	Define **radioactivity**. Complete the nuclear change and name the particle emitted: ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th$ + ______	**3**	**OR**	What is **carbon dating**? Carbon-14 decays into nitrogen-14; state what happens to the number of protons in the nucleus and name the type of particle emitted.	**3**
2(x)	What are the **lanthanides and actinides**? To which block do they belong, and why are they shown in two separate rows at the bottom of the periodic table?	**3**	**OR**	Predict, with a reason for each, how the following change as you go **down** Group 1 (alkali metals): (a) reactivity (b) density (c) melting point.	**3**
2(xi)	What are **dipole-dipole forces**? Between what kind of molecules do they act, and how do they arise? Give one example.	**3**	**OR**	Explain why **solid** sodium chloride does not conduct electricity but **molten** (or aqueous) sodium chloride does. Do aqueous acids conduct electricity? Give a reason.	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	(a) Define relative atomic mass. Naturally occurring uranium has three isotopes: U-238 (99.27%, mass 238), U-235 (0.72%, mass 235) and U-234 (0.006%, mass 234). Calculate the relative atomic mass of uranium. [3.5] (b) Write the balanced nuclear equation for the alpha decay of U-238 into thorium-234 and name the particle emitted. [1.5]	**5**	**OR**	(a) Define relative atomic mass and explain why carbon-12 is chosen as the standard. Calculate the relative atomic mass of carbon from its isotopes C-12 (98.8%), C-13 (1.1%) and C-14 (0.009%). [3.5] (b) Carbon-14 decays into nitrogen-14. Complete ${}^{14}_{6}C \rightarrow {}^{14}_{7}N$ + _____ and state one use of carbon-14. [1.5]	**5**
3(ii)	A compound contains 54.5% carbon, 9.1% hydrogen and 36.4% oxygen by mass; its relative molecular mass is 88 amu. (C = 12, H = 1, O = 16) (a) Determine its empirical formula. [3] (b) Determine its molecular formula. [2]	**5**	**OR**	A hydrocarbon contains 82.7% carbon and 17.3% hydrogen by mass; its relative molecular mass is 58 amu. (C = 12, H = 1) (a) Determine its empirical formula. [3] (b) Determine its molecular formula. [2]	**5**
3(iii)	Nitrogen and hydrogen react to form ammonia: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$. Using the bond energies N≡N = 945, H–H = 436 and N–H = 391 kJ/mol, calculate ΔH for the reaction and state whether it is exothermic or endothermic. Show all steps. [5]	**5**	**OR**	(a) Define activation energy. Using collision theory, explain the difference between an effective and an ineffective collision. [3] (b) Draw and label a reaction-pathway (energy) diagram for an **exothermic** reaction, showing reactants, products, activation energy with and without a catalyst, and ΔH. State the sign of ΔH. [2]	**5**
3(iv)	Propane burns completely: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$. A cylinder releases 4.4 g of propane. (C = 12, H = 1, O = 16, $N_A = 6.022\times10^{23}$) (a) Moles of propane. [1] (b) Moles of O_2 required. [1] (c) Mass of CO_2 produced. [2] (d) Number of water molecules produced. [1]	**5**	**OR**	Using dot-and-cross (electron) diagrams, show the bonding in each molecule and state the number of bonding pairs and lone pairs on the central atom: (a) methane, CH_4 [1.5] (b) water, H_2O [1.5] (c) carbon dioxide, CO_2 [2]. (H = 1, C = 6, O = 8)	**5**

Invigilator's Signature: ___________________ Candidate's Roll No.: ___________________

Chapters Covered: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)

— End of Question Paper —